Week 1: Data Gathering, Cleaning, and Preprocessing

Data Science Internship Task Report

1. Objective

The objective of this task was to collect a publicly available dataset, explore its structure and quality, identify missing, inconsistent, or abnormal values, apply appropriate cleaning and preprocessing steps, and document the methodology for downstream analysis.

2. Dataset Source and Overview

Dataset: Wine recognition dataset, a public benchmark associated with the UCI Machine Learning Repository and available through scikit-learn. The dataset contains chemical measurements of wine samples belonging to three classes.

Attribute	Value
Rows / observations	178
Columns (including target)	14
Predictor variables	13
Target variable	wine_class (3 classes)
Missing cells detected	0
Duplicate rows detected	0

3. Initial Data Exploration

The dataset contains 178 observations and 13 numeric measurement features plus the target label. Numeric summaries, including minimum, maximum, mean, and quartiles, were reviewed. Class distribution was checked to understand the target balance.

Wine class	Count
class_1	71
class_0	59
class_2	48

4. Data Quality Issues and Treatment

Missing values: No missing cells were found in the source dataset. Therefore, no imputation or row deletion was necessary.

Duplicate records: No exact duplicate rows were detected, so no duplicate removal was required.

Outliers: Potential outliers were flagged using the 1.5 × IQR rule for each numeric feature. These are statistical flags rather than confirmed errors. Since the values represent real chemical measurements and may be scientifically meaningful, flagged values were retained rather than deleted or capped.

Feature	IQR outlier flags
alcohol	0

malic_acid	3
ash	3
alcalinity_of_ash	4
magnesium	4
total_phenols	0
flavanoids	0
nonflavanoid_phenols	0
proanthocyanins	2
color_intensity	4
hue	1
od280/od315_of_diluted_wines	0
proline	0

Inconsistent values: The dataset contains numeric measurements and a class label. No invalid text entries or inconsistent category spellings were identified. Integer target codes were mapped to descriptive class names.

5. Data Type Conversion and Preprocessing

- All 13 measurement columns were explicitly converted to numeric float64 values.
- The target column was renamed to wine_class and converted to a categorical data type.
- No normalization or standardization was applied to the saved dataset. Scaling is model-dependent and should be fitted on training data only to avoid data leakage.
- The cleaned dataset was exported as CSV without an extra index column.

6. Validation After Cleaning

Post-cleaning validation confirmed 178 rows and 14 columns, 0 missing cells, and 0 duplicate rows. Numeric features are stored as float64 and wine_class is categorical. All source observations were preserved.

7. Summary and Conclusion

The dataset was successfully sourced, explored, validated, and prepared for further analysis. No missing values or duplicate records required removal. Potential outliers were documented and retained because there was no evidence that they were data-entry errors. Numeric types and the target category were standardized, and the cleaned dataset was exported as CSV. The dataset is suitable for exploratory data analysis and can be scaled within a downstream modeling pipeline.

8. Dataset References

UCI Machine Learning Repository, Wine dataset: https://archive.ics.uci.edu/dataset/109/wine

Scikit-learn documentation, load_wine: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_wine.html

Dataset accessed through sklearn.datasets.load_wine(as_frame=True).